

Sketching Graphs of Quadratic Functions

Jonathan R. Bacolod

A **quadratic function** has the general form

$$y = ax^2 + bx + c, \text{ where } a \neq 0.$$

Parabola

The graph of a quadratic function is called a **parabola**.

$$y = ax^2 + bx + c$$

Direction of Opening

If $a > 0$, the parabola **opens upward**.

If $a < 0$, the parabola **opens downward**.

$$a > 0 \Rightarrow \text{opens upward}$$

$$a < 0 \Rightarrow \text{opens downward}$$

Vertex

The **vertex** is the highest or lowest point of the parabola.

If the parabola opens upward, the vertex is the lowest point.

If the parabola opens downward, the vertex is the highest point.

Axis of Symmetry

The **axis of symmetry** is the vertical line passing through the vertex.

$$x = -\frac{b}{2a}$$

The graph is always symmetric about its axis of symmetry.

y -Intercept

The **y -intercept** is found by letting $x = 0$.

For

$$y = ax^2 + bx + c$$

let $x = 0$ to find the y -intercept.

x -Intercepts

The x -**intercepts**, if they exist, are the zeros of the quadratic function.

They are the points where the graph crosses or touches the x -axis.

$$y = 0$$

Sketching a Parabola

A graph can be sketched by identifying the:

vertex

axis of symmetry

intercepts

additional symmetric points

Neat Graph

A neat sketch should include:

- labeled axes

- intercepts

- vertex

- direction in which the parabola opens

Examples

Example 1

Sketch the graph of

$$y = x^2 - 4x + 3.$$

Label the vertex, axis of symmetry, intercepts, and direction of opening.

Example 2

Sketch the graph of

$$y = x^2 + 2x - 8.$$

Label the vertex, axis of symmetry, intercepts, and direction of opening.

Example 3

Sketch the graph of

$$y = -x^2 + 6x - 5.$$

Label the vertex, axis of symmetry, intercepts, and direction of opening.

Practice Exercises

Practice 1

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 - 4x + 3$$

Practice 2

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 + 2x - 8$$

Practice 3

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 + 6x - 5$$

Practice 4

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = 2x^2 - 8x + 6$$

Practice 5

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -2x^2 - 4x + 6$$

Practice 6

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 + 6x + 5$$

Practice 7

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 - 10x + 21$$

Practice 8

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 + 4x$$

Practice 9

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = 3x^2 - 12x + 9$$

Practice 10

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 - 2x + 8$$

Activities

Activity 1

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 - 6x + 5$$

Activity 2

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 + 4x + 3$$

Activity 3

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 + 8x - 12$$

Activity 4

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = 2x^2 - 4x - 6$$

Activity 5

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 - 8x + 12$$

Activity 6

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 + 8x + 15$$

Activity 7

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 + 2x + 3$$

Activity 8

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = 3x^2 + 6x - 9$$

Activity 9

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 - 6x + 7$$

Thank you!